



ST JOSEPH'S UNIVERSITY, BENGALURU -27

B.Sc (MATHEMATICS) - III SEMESTER

ACTIVITY I

MT326-MATHEMATICS III-Calculus and Differential Equation I

Answer ALL of the following.

1. Use the following theorem, discussed in class, to compute the upper and lower integrals of each function. Hence determine whether the function is integrable on the given interval.

Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let (P_n) be a sequence of partitions of $[a, b]$ such that $\|P_n\| \rightarrow 0$. Then

$$\lim_{n \rightarrow \infty} U(f, P_n) = \int_a^b f(x) dx, \quad \lim_{n \rightarrow \infty} L(f, P_n) = \int_a^b f(x) dx.$$

- (a) $f(x) = x + 1$ on $[0, 2]$.
- (b) $f(x) = x^2 + 1$ on $[1, 3]$.
- (c) $f(x) = \cos x$ on $\left[0, \frac{\pi}{2}\right]$.
- (d) $f(x) = x^3$ on $[0, 1]$.
- (e) $f(x) = x^9$ on $[1, 2]$.

Hint for part (c). Use the uniform partition

$$P_n = \left\{ 0, \frac{\pi}{2n}, \frac{2\pi}{2n}, \dots, \frac{n\pi}{2n} \right\}.$$

You may use the trigonometric formulas

$$\sum_{k=0}^{n-1} \cos(k\theta) = \frac{\sin\left(\frac{n\theta}{2}\right) \cos\left(\frac{(n-1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

and

$$\sum_{k=1}^n \cos(k\theta) = \frac{\sin\left(\frac{n\theta}{2}\right) \cos\left(\frac{(n+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}.$$

Hint for part (d). You may use

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Hint for part (e). Instead of using equal subintervals, try

$$P_n = \{a, ar, ar^2, \dots, ar^n\}, \quad r = \left(\frac{b}{a}\right)^{1/n},$$

for an interval $[a, b]$, where $a > 0$.

2. Let $c \in (0, 1)$, and define $f : [0, 1] \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 1, & \text{if } x = c, \\ 0, & \text{if } x \neq c. \end{cases}$$

Show that f is integrable on $[0, 1]$ and that

$$\int_0^1 f(x) dx = 0.$$

3. Find the general solution of the differential equation $(x^2 - y^2) \frac{dy}{dx} + (y^2 + x^2) = 0$.
4. For the differential equation $\frac{dy}{dx} = (1+x)(1+y^2)$ find the integral curve that passes through the point $(0, 1)$.
5. Show that the family of parabolas $y^2 = 4a(x+a)$ is self orthogonal.
6. Find the orthogonal trajectories of the family $r = a(1 - \cos\theta)$

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